CUBAN
NATIONAL CHEMISTRY CONTEST 2022

ANSWERS TO THE PROBLEMS

"You should never be afraid of what you're doing, when it's right."

Marie Curie

Ministerio de Educación

República de Cuba

FACULTAD DE QUÍMICA
UNIVERSIDAD DE LA HABANA

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Revision

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CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6{,}0221 \cdot 10^{23}\ mol^{-1}$	Equation of the ideal gas	$p \cdot V = n \cdot R \cdot T$
Gases' constant	$R = 8{,}3145\ J \cdot K^{-1} \cdot mol^{-1}$	First Principle of Thermodynamics	$Q = \Delta E + W$
Zero on the Celsius scale	$273{,}15\ K$	Gibbs' free energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485\ C \cdot mol^{-1}$	Nernst's equation	$E = E° - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{red}}{c_{ox}}$
Ionic product of water at 25 °C	$K_W = 1{,}00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp(-\frac{E_A}{R \cdot T})$
Standard conditions	$p° = 100\ kPa$ $c° = 1\ mol \cdot L^{-1}$	$\Delta G° = -R \cdot T \cdot \ln K° = -z \cdot F \cdot \Delta E°$	

$1\ J = 1\ C \cdot 1\ V = 1\ kPa \cdot 1\ L = 1\ W \cdot 1\ s$

$1\ C = 1\ A \cdot 1\ s$

PERIODIC TABLE OF ELEMENTS

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H 1.008																	2 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm -	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
89 Ac -	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np -	94 Pu -	95 Am -	96 Cm -	97 Bk -	98 Cf -	99 Es -	100 Fm -	101 Md -	102 No -	103 Lr -

Problem 1. Miscellanea (16 points)

1.1	…	Total	Total
1	idem	16	16

Select the correct answer in each case.

1.1. In the sequence of ions S^{2-}, Br^-, Ag^+, Cu^{2+}, Mg^{2+}, the one with the largest ionic radius is:

A) S^{2-}	B) Ag^+	C) Br^-	D) Cu^{2+}

1.2. Group of chemical species with the same bond order:

A) NO^+, O_2^{2+}, O_2^-, N_2, CN^-	B) N_2^{2-}, O_2^{2-}, NO^-, O_2, CN^+
C) NO, O_2^+, N_2^-, NO^{2+}, CN	D) none of the previous

1.3. Which molecule is polar and has a sp^2-hybridized central atom?

A) BCl_3	B) $HClO_2$	C) $POCl_3$	D) H_2CO_3

1.4. The covalent character of the bond present in the halides LiCl, $BeCl_2$, BCl_3 and CCl_4 increases in the order:

A) $LiCl < BeCl_2 < BCl_3 < CCl_4$	B) $CCl_4 < BCl_3 < BeCl_2 < LiCl$
C) $LiCl < BeCl_2 < BCl_3 > CCl_4$	D) $LiCl < BCl_3 < BeCl_2 < CCl_4$

1.5. Which molecule can act as a Lewis acid?

A) $SnCl_4$	B) SiF_4	C) CCl_4	D) SF_4

1.6. If the enthalpy of formation of liquid water is -286 kJ/mol and that of ice is -292 kJ/mol, how much energy is released in the combustion of 1 g of ice?

A) 6 kJ	B) 16.2 kJ	C) 0	D) 333 J

1.7. Which oxide has amphoteric acid-base behavior?

A) CaO	B) ZnO	C) CO_2	D) SiO_2

1.8. The bromination of acetone varies its initial rate from 7.6 mol/(L·s) to 15.2 mol/(L·s) when the pH decreases from 1.3 to 1, at constant the concentrations of the other reagents. Knowing that pH = - log(H^+), what is the reaction order respect to H^+?

A) −1	B) 1	C) 0,5	D) 2

1.9. Which of the following chemical species is paramagnetic?

A) H_2	B) HHe^+	C) He_2^+	D) He_2

1.10. In which of the oxo-acids below the oxidation state of phosphorus isn't 5+?

A) $H_4P_2O_7$	B) H_3PO_4	C) $H_4P_2O_6$	D) $H_5P_3O_{10}$

1.11. In the reaction between sodium amide and sodium nitrate, in liquid ammonia under controlled conditions, represented by the following unbalanced chemical equation:

$$NaNH_2(solv) + NaNO_3(solv) \rightarrow NaN_3(s) + NaOH(solv) + NH_3(l)$$

for every mole of sodium amide that is consumed, the following amount is formed:

A) 2 mol of NaN_3	B) 1 mol of NaN_3
C) 1/2 mol of NaN_3	D) 1/3 mol of NaN_3

1.12. Which molecule has T-shaped geometry?

A) $SOCl_2$	B) ICl_3	C) PBr_3	D) $COCl_2$

1.13. For the reaction between Cl_2 and NO, producing NOCl, it is known that at low concentrations of the reducing agent, the experimental rate law reduces to v = $k \cdot c^2(NO) \cdot c(Cl_2)$. What could be the complete expression (not simplified) that is valid for all cases?

A) $v = \frac{k_2 k_1 c^3(NO)c(Cl_2)}{k_{-1}+k_2 c(NO)}$	B) $v = \frac{k_2 k_1 c^2(NO)c(Cl_2)}{k_{-1}+k_2 c(NO)}$
C) $v = \frac{k_2 k_1 c^2(NO)c(Cl_2)}{k_{-1}+k_2 c(Cl_2)}$	D) $v = \frac{k_2 k_1 c^3(NO)c(Cl_2)}{k_{-1}+k_2 c(Cl_2)}$

1.14. A gas occupies a volume of 9.23 L at 345 K and 1.40 atm. The volume that this gas will occupy at 525 K and 3.20 atm will be:

A) 5.65 L	B) 6.14 L	C) 321 mL	D) 11.9 L

1.15. Which of the following ions doesn't have a completely filled d orbital?

A) Ag^+	B) Ga^{3+}	C) Cu^{2+}	D) Zn^{2+}

1.16. A 1.50 g sample of a silver-containing mineral was dissolved and all silver was converted to 0.124 g of Ag_2S. The percentage of silver in the ore sample was:

A) 10.8%	B) 6.41%	C) 8.27%	D) 7.20%

Problem 2. Finding the elements (20 points)

2.1	2.2	2.3	2.4	Total	Total
8	2	7	3	20	20

A volatile liquid with formula $\mathbf{X}\mathbf{Y}_2$ is produced by passing a current of the hydride $\mathbf{X}H_4$ over the simple substance of the element **Y**, also forming the gaseous compound with an unpleasant odor $H_2\mathbf{Y}$. Combustion of 1 mol of $\mathbf{XY}_2$ in an oxygen atmosphere produces 172.1 g of the oxides $\mathbf{X}O_2$ and $\mathbf{Y}O_2$. The mass ratio of element **Y** to element **X** in $\mathbf{XY}_2$ is 5.339. Reaction of $\mathbf{XY}_2$ with ammonia, in the presence of a catalyst, gives as the only products $H_2\mathbf{Y}$ and an ammonium salt containing 42.1% **Y** and a molar ratio of element **X** to nitrogen of 1:2.

2.1 Identify elements **X** and **Y**. Justify numerically.

$\mathbf{XY}_2 + 3\,O_2 = \mathbf{X}O_2 + 2\,\mathbf{Y}O_2$ (1 mark)

Combustion of 1 mol of $\mathbf{XY}_2$ produces 1 mol of $\mathbf{X}O_2$ and 2 mol of $\mathbf{Y}O_2$

$m(\mathbf{X}O_2) = 1\cdot(M(\mathbf{X}) + 32)$ $m(\mathbf{Y}O_2) = 2\cdot(M(\mathbf{Y}) + 32)$ (1 mark)

$m(\mathbf{X}O_2) + m(\mathbf{Y}O_2) = (M(\mathbf{X}) + 32) + 2\cdot(M(\mathbf{Y}) + 32)$ (1 mark)

$172.1 = M(\mathbf{X}) + 2M(\mathbf{Y}) + 96$

$M(\mathbf{X}) + 2M(\mathbf{Y}) = M(\mathbf{XY}_2) = 76.1$ g/mol (2 marks)

On the other hand, it is found that the **Y**:**X** mass ratio in $\mathbf{XY}_2$ is 5.339, which can be expressed as a function of molar masses as:

$2M(\mathbf{Y})/M(\mathbf{X}) = 5{,}339$

Combining with the previous equation, it is obtained that:

$M(\mathbf{X}) + 5.339M(\mathbf{X}) = 76.1$ (1 mark)

$M(\mathbf{X}) = 12.0$ g/mol; this atomic weight corresponds to carbon (1 mark)

$M(\mathbf{Y}) = (76.1 – M(\mathbf{Y}))/2 = 32.05$; this atomic weight corresponds to sulphur (1 mark)

Total (8 marks)

2.2 Illustrate the Lewis structure of $\mathbf{XY}_2$ molecule. Why does this compound have a relatively low boiling point?

S=C=S (1 mark)

Because the dipole moment of CS_2 is zero, the interactions between the molecules are very weak (fluctuating dipole-induced dipole), and its boiling temperature is very low (if compared to other substances of similar molar mass). (1 mark)

Total (2 marks)

2.3 Write the formula and the name of the salt obtained in the reaction with ammonia. Show the calculations that support your reasoning.

$$0.421 = \frac{m(S)}{m(salt)} = \frac{n(S)*32.06}{m(salt)}$$

$$m(salt) = \frac{n(S)*32.06}{0.421} = 76.15 * n(S)$$ (1 mark)

Taking into account that n(N) = 2 n(C), then there is a unit of molar mass $12.01 + 14.01 * 2$ = 40.03 g/mol. For the case of n(S) = 1, M(salt) = 76.15, subtracting the molar mass of sulfur and the unit of 1 carbon and two nitrogens, leaves a molar mass value of:

$$M(residue, H) = 76.15 - (32.06 + 40.03) = 4.06\ g/mol$$

This corresponds to 4 hydrogen atoms, or one ammonium (NH_4) unit, this leaves the global formula N_2H_4CS as the only possibility (4 marks)

The formula of the salt is NH_4SCN (1 mark)

The name is ammonium thiocyanate (1 mark)

Total (7 marks)

2.4 Write the balanced equations corresponding to the processes described above.

CH_4 (g) + ½ S_8 (s) = CS_2 (l) + 2 H_2S (g) (1 mark)

CS_2 (l) + 3 O_2 (g) = CO_2 (g) + 2 SO_2 (g) (1 mark)

CS_2 (l) + 2 NH_3 (g) = NH_4SCN (s) + H_2S (g) (1 mark)

Total (3 marks)

Problem 3. Bauxite (20 points)

3.1	3.2	3.3	3.4	Total	Total
6	25	5	4	40	20

Bauxite is the most used mineral to obtain aluminum. In addition to aluminum, the main metals contained in bauxite are iron, silicon and titanium. Bauxite composition varies depending on where it is mined. In a mine of this mineral, in Greece, it was found that when water was added to the product a solid residue made up of Al_2O_3, SiO_2, Fe_2O_3 and TiO_2 remained undissolved. This residue, by subsequent analysis, was found to have an approximate composition $m\mathrm{Al_2O_3}\cdot n\mathrm{SiO_2}\cdot p\mathrm{Fe_2O_3}\cdot q\mathrm{TiO_2}$, where occur the relationships $m/p = p/q$ and $q/n = n/p$.

The addition of 1.215 g of bauxite ore, extracted directly from the mine, to a diluted HNO_3 solution, gives a solid residue of 0.0820 g. On the other hand, by taking the same mass of bauxite and adding a NaOH solution, the mass of residue that remains at the bottom has a mass of 0.3735 g. It is known that the Fe^{3+} ion in the presence of NaOH forms a precipitate with the formula FeO(OH).

3.1 Write balanced equations of the reactions described. (Note that under these conditions neither SiO_2 nor TiO_2 react)

Al_2O_3 (s) + 6 HNO_3 (aq) = 2 $Al(NO_3)_3$ (aq) + 3 H_2O (1 mark)
Fe_2O_3 (s) + 6 HNO_3 (aq) = 2 $Fe(NO_3)_3$ (aq) + 3 H_2O (1 mark)
Al_2O_3 (s) + 6 NaOH (aq) = 2 Na_3AlO_3 (aq) + 3 H_2O (2 marks)
(It's also acceptable: Al_2O_3 (s) + 2 NaOH (aq) = 2 $NaAlO_2$ (aq) + H_2O)
Fe_2O_3 (s) + H_2O $\xRightarrow{NaOH}$ 2 FeO(OH) (s) (2 marks)

Total (6 marks)

3.2 Calculate the mass composition of bauxite ore.

In 1.215 g of bauxite ore when dissolved in HNO_3, a residue of 0.0820 g composed of SiO_2 and TiO_2 remains.
In 1.215 g of bauxite ore when dissolved in NaOH, a residue of 0.3735 g is left, composed of SiO_2, TiO_2 and FeO(OH).
m(FeO(OH)) = 0.3735 – 0.0820 = 0.2915 g (2 marks)
$n(\mathrm{FeO(OH)}) = \frac{m(\mathrm{FeO(OH)})}{M(\mathrm{FeO(OH)})} = \frac{0.2915\mathrm{g}}{88.86\ \mathrm{g/mol}}$ = 0.00328 mol
n(Fe_2O_3) = ½ ∗ 0.00328mol = 0.00164 mol (2 marks)
$m(\mathrm{Fe_2O_3}) = n(\mathrm{Fe_2O_3}) * M(\mathrm{Fe_2O_3}) = 0.00164\mathrm{mol} * 159.7\mathrm{g/mol}$
m(Fe_2O_3) = 0.2619 g (1 mark)
$\omega(\mathrm{Fe_2O_3})\cdot 100 = \frac{0.2619\mathrm{g}}{1.215\mathrm{g}} * 100 = 21.56\%$ (1 mark)

From the molar ratio of the residue q/n = n/p, it follows that:

$\frac{n(TiO_2)}{n(SiO_2)} = \frac{n(SiO_2)}{n(Fe_2O_3)}$, and therefore $n^2(SiO_2) = n(Fe_2O_3) * n(TiO_2)$ (1 mark)

$n^2(SiO_2) = 0.00164 * \frac{m(TiO_2)}{79.87}$

$\frac{79.87}{0.00164} * n^2(SiO_2) = m(TiO_2) = 0.0820 - m(SiO_2)$ (2 marks)

$48701n^2(SiO_2) = 0.0820 - M(SiO_2) * n(SiO_2) = 0.0820 - 60.09n(SiO_2)$

$48701n^2(SiO_2) + 60.09n(SiO_2) - 0.0820 = 0$ (3 marks)

$n(SiO_2) = \frac{-60.09 \pm \sqrt{60.09^2 + 4*48701*0.0820}}{2*48701}$

$n(SiO_2) = 8.2 \cdot 10^{-4}$ mol (2 marks)

$m(SiO_2) = n(SiO_2) * M(SiO_2) = 8.2 \cdot 10^{-4} mol * 60.09 g/mol = 0.0493$ g (1 mark)

$\omega(SiO_2) \cdot 100 = \frac{0.0493g}{1.215g} * 100 = 4.06\%$ (1 mark)

$n(TiO_2) = \frac{n^2(SiO_2)}{n(Fe_2O_3)} = \frac{(8.2 \cdot 10^{-4})^2}{0.00164} = 4.1 \cdot 10^{-4}$ mol (2 marks)

$m(TiO_2) = n(TiO_2) * M(TiO_2) = 4.1 \cdot 10^{-4} mol * 79.87 g/mol = 0.0327$ g (1 mark)

$\omega(TiO_2) \cdot 100 = \frac{0.0327g}{1.215g} * 100 = 2.69\%$ (1 mark)

From the molar ratio of the residue m/p = p/q, it follows that:

$\frac{n(Al_2O_3)}{n(Fe_2O_3)} = \frac{n(Fe_2O_3)}{n(TiO_2)}$, and therefore $n(Al_2O_3) = \frac{n^2(Fe_2O_3)}{n(TiO_2)} = \frac{(0.00164)^2}{8.2 \cdot 10^{-4}} = 0.00656$ mol (2 marks)

$m(Al_2O_3) = n(Al_2O_3) * M(Al_2O_3) = 0.00656 mol * 101.96 g/mol = 0.6689 g$ (1 mark)

$\omega(Al_2O_3) \cdot 100 = \frac{0.6689g}{1.215g} * 100 = 55.05\%$ (1 mark)

ω(soluble residue)·100 = 100 – (55.05 + 2.69 + 4.06 + 21.56) = 16.64% (1 mark)

Total (25 marks)

3.3 Propose a formula for the residue $m Al_2O_3 \cdot n SiO_2 \cdot p Fe_2O_3 \cdot q TiO_2$, taking the smallest coefficient to be equal to 1.

The smallest coefficient corresponds to TiO_2, which has the least amount of substance in the mineral. Dividing the amounts of substances of all oxides by that of TiO_2, it is obtained:

$m = \frac{0.00656 \text{ mol}}{4.1 \cdot 10^{-4} \text{ mol}} = 16$; $n = \frac{8.2 \cdot 10^{-4} \text{ mol}}{4.1 \cdot 10^{-4} \text{ mol}} = 2$;

$p = \frac{0.00164 \text{ mol}}{4.1 \cdot 10^{-4} \text{ mol}} = 4$; $q = \frac{4.1 \cdot 10^{-4} \text{ mol}}{4.1 \cdot 10^{-4} \text{ mol}} = 1$ (1 mark each)

The formula of the residue is:

$16\ Al_2O_3 \cdot 2\ SiO_2 \cdot 4\ Fe_2O_3 \cdot TiO_2$ (1 mark)

Total (5 marks)

3.4 What is the maximum mass of Al that can be obtained from 1 t of this bauxite ore, assuming 85% yield in the entire metal separation and purification process.
If you could not solve part 3.2, consider $\omega(Al_2O_3)$ = 50%

$\omega(Al_2O_3)\cdot 100 = 55.05\%$	
In 1 t of bauxite ore there are 550.5 kg of Al_2O_3	(1 mark)
$m_{max}(Al) = \frac{m(Al_2O_3) * M(Al) * 2}{M(Al_2O_3)} * \nu = \frac{550.5\ kg * 26.98\ g/mol * 2}{101.96\ g/mol} * 0.85 = 247.6\ kg$	(3 marks)
Total	(4 marks)

Using the value of $\omega(Al_2O_3)$ = 50%, a mass of aluminum of 224.9 kg is obtained.

Problem 4. Kinetics (14 points)

4.1	4.2	4.3	Total	Total
7	1	6	14	14

Nitramide slowly decomposes in aqueous solution according to the reaction:
$NO_2NH_2(aq) \rightarrow N_2O(g) + H_2O$

The kinetic law, determined experimentally, is the following:
$\frac{d(N_2O)}{dt} = k\frac{[NO_2NH_2]}{[H_3O^+]}$

4.1 Which of the following mechanisms is the most appropriate for the interpretation of the experimental kinetic law?

Mechanism 1

$NO_2NH_2 + H_3O^+ \underset{k_{-1}}{\overset{k_1}{\rightleftarrows}} NO_2NH_3^+ + H_2O$ (fast equilibrium)

$NO_2NH_3^+ \xrightarrow{k_2} N_2O + H_3O^+$ (limiting step)

Mechanism 2

$NO_2NH_2 + H_2O \underset{k_{-3}}{\overset{k_3}{\rightleftarrows}} NO_2NH^- + H_3O^+$

$NO_2NH^- \xrightarrow{k_4} N_2O + OH^-$ (limiting step)

$H_3O^+ + OH^- \xrightarrow{k_5} H_2O$ (very fast reaction)

For mechanism 1

$k_1[NO_2NH_2][H_3O^+] = k_{-1}[NO_2NH_3^+][H_2O]$ (1 mark)

$[NO_2NH_3^+] = \frac{k_1[NO_2NH_2][H_3O^+]}{k_{-1}[H_2O]}$ (1 mark)

$\frac{d(N_2O)}{dt} = k_2[NO_2NH_3^+] = \frac{k_1k_2[NO_2NH_2][H_3O^+]}{k_{-1}[H_2O]}$ (1 mark)

For mechanism 2

$k_3[NO_2NH_2][H_2O] = k_{-3}[NO_2NH^-][H_3O^+]$ (1 mark)

$[NO_2NH^-] = \frac{k_3[NO_2NH_2][H_2O]}{k_{-3}[H_3O^+]}$ (1 mark)

$\frac{d(N_2O)}{dt} = k_4[NO_2NH^-] = \frac{k_3k_4[NO_2NH_2][H_2O]}{k_{-3}[H_3O^+]}$ (1 mark)

Therefore, of the two mechanisms, the only one that complies with the relationship given in the rate law is mechanism 2. (1 mark)

Total (7 marks)

4.2 Find the relationship between the experimental constant and the rate constants of the individual steps in the chosen mechanism.

$K = \frac{k_3 k_4}{k_{-3}}[H_2O]$	(1 mark)
Total	(1 mark)

4.3 If the rate of the reaction is increased five-fold when increasing the temperature by 15°C from 298 K, calculate the activation energy of this reaction.

$k = A * \exp\left(-\frac{\Delta H^*}{RT}\right)$	
$\frac{v_2}{v_1} = \frac{k_2}{k_1} = \frac{\exp\left(-\frac{\Delta H^*}{RT_2}\right)}{\exp\left(-\frac{\Delta H^*}{RT_1}\right)} = \exp\left(-\frac{\Delta H^*}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)\right)$	(2 marks)
$\ln\left(\frac{v_2}{v_1}\right) = -\frac{\Delta H^*}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$	(1 mark)
$\Delta H^* = -R\left(\frac{1}{T_2} - \frac{1}{T_1}\right)^{-1} * \ln\left(\frac{v_2}{v_1}\right)$	(1 mark)
$\Delta H^* = -8.3145\ J/(mol \cdot K)\left(\frac{1}{313.15} - \frac{1}{298.15}\right)^{-1} * \ln(5) = 83.3\ kJ/mol$	(2 marks)
Total	(6 marks)

Problem 5. Sulfur chemistry (30 points)

5.1	5.2	5.3	5.4	5.5	5.6	Total	Total
9	23	4	4	15	5	60	30

As for other non-metals, the existence of a variety of compounds with sulfur-halogen bonds has been demonstrated. Many of them play a fundamental role in the synthesis of organic or inorganic substances. The chemical synthesis of some sulfur halides or oxohalides is described below:

i) Sulfur hexafluoride, SF_6, is a gas at STP obtained by combustion of elemental sulfur in a fluorine atmosphere.

ii) S_2F_{10} can be obtained in the liquid state under similar conditions to the previous ones, but only if temperature and pressure are carefully controlled.

iii) Sulfur tetrafluoride vapors, SF_4, are obtained together with SF_6, as products of the S_2F_{10} disproportionation when heated to 150 °C.

iv) At STP, the gaseous compounds SF_5Cl and SF_5Br are formed by the treatment of the S_2F_{10} with chlorine and bromine, respectively.

v) SOF_4 vapors are obtained by air oxidation of sulfur tetrafluoride.

vi) Sulphuryl fluoride, SO_2F_2, is a gas at STP formed by the reaction of sulfur dioxide with fluorine.

vii) S_2Cl_2 is a liquid substance at STP, with a structure like hydrogen peroxide. This compound is obtained from the combustion of sulfur in chlorine atmosphere.

viii) Liquid sulfur dichloride, SCl_2, is formed by post-treatment of S_2Cl_2 with chlorine.

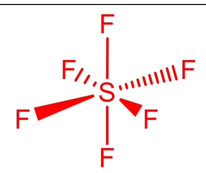
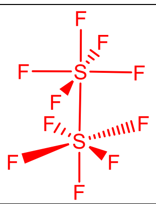
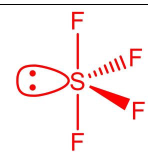
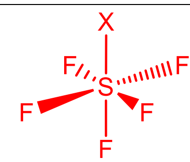
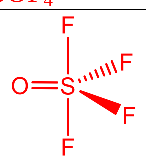
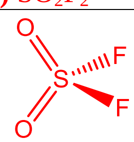
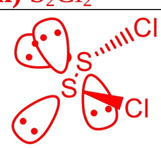
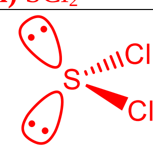
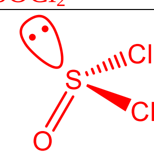
ix) Liquid thionyl chloride, $SOCl_2$, is industrially prepared by the reaction of sulfur trioxide with sulfur dichloride in chlorine atmosphere.

5.1 Write the equations for the nine reactions described above.

i) SF_6: $S_8(s) + 24\ F_2(g) \triangleq 8\ SF_6(g)$	(1 mark)
ii) S_2F_{10}: $S_8(s) + 20\ F_2(g) \triangleq 4\ S_2F_{10}(l)$	(1 mark)
iii) SF_4: $S_2F_{10}(g) \overset{150^{\circ}C}{=} SF_6(g) + SF_4(g)$	(1 mark)
iv) SF_5X (X = Cl, Br): $S_2F_{10}(g) + X_2(g) = 2\ SF_5X(g)$	(1 mark)
v) SOF_4: $2SF_4(g) + O_2(g) = 2SOF_4(g)$	(1 mark)
vi) SO_2F_2: $SO_2(g) + F_2(g) = SO_2F_2(g)$	(1 mark)
vii) S_2Cl_2: $S_8(s) + 4\ Cl_2(g) \triangleq 4\ S_2Cl_2(l)$	(1 mark)
viii) SCl_2: $S_2Cl_2(l) + Cl_2(g) = 2\ SCl_2(l)$	(1 mark)
ix) $SOCl_2$: $SO_3(g) + 2SCl_2(l) + Cl_2(g) = 3\ SOCl_2(l)$	(1 mark)
Total	(9 marks)

5.2 What is the most probable geometry adopted by the domains (bonding or non-bonding pairs of electrons) around the sulfur atoms in each of the obtained compounds? Draw the corresponding structures according to the VSEPR theory. What molecules have zero net dipole moment?

NOTE: For each compound, write in the corresponding box the chemical structure, the type of geometry and whether or not the molecule has a zero net dipole moment (write Yes or No when appropriate).

Chemical formula	**i)** SF_6	**ii)** S_2F_{10}	**iii)** SF_4	**iv)** SF_4X (X = Cl, Br)
Chemical structure				
	(1 mark)	(1,5 mark)	(1 mark)	(1 mark)
Geometry	*octahedral*	*octahedral*	*trigonal bipyramid*	*octahedral*
	(1 mark)	(1 mark)	(1 mark)	(1 mark)
$\mu = 0$	*Yes*	*Yes*	*No*	*No*
	(0,5 mark)	(0,5 mark)	(0,5 mark)	(0,5 mark)
v) SOF_4	**vi)** SO_2F_2	**vii)** S_2Cl_2	**viii)** SCl_2	**ix)** $SOCl_2$
				
(1 mark)	(1 mark)	(1 mark)	(1 mark)	(1 mark)
trigonal bipyramid	*tetrahedral*	*tetrahedral*	*tetrahedral*	*tetrahedral*
(1 mark)	(1 mark)	(1 mark)	(1 mark)	(1 mark)
No	*No*	*No*	*No*	*No*
(0,5 mark)	(0,5 mark)	(0,5 mark)	(0,5 mark)	(0,5 mark)
Total				(23 marks)

5.3 The higher reactivity of the S_2F_{10} compared to sulfur hexafluoride makes it a suitable starting material for synthesis of other halogenated sulfur compounds. What features of their structures lead to this difference in reactivity?

The reactions of SF_6 and S_2F_{10} in which more complex sulfur halides are obtained, involve the breaking of one of their bonds to form a new bond. The difference in reactivity of both sulfur compounds depends mainly on the strength of their bonds: if the molecule has strong bonds, the reactions will proceed slowly. Sulfur hexafluoride presents six strong S-F bonds (polar covalent character and shorter bond distance), while S_2F_{10}, in addition to the ten S-F bonds, also presents a weaker S-S bond (non-polar covalent character, and longer bond distance). For this reason, S_2F_{10}, with a weaker S-S bond than the S-F bonds of sulfur hexafluoride, is a more reactive substance. (4 marks)

Total (4 marks)

5.4 Compare the S-X bond length and X-S-F bond angle of the SF_5X compounds, X = F, Cl, Br.

NOTE: Write the chemical formula of the SF_5X compounds in such a way that there is an increasing order of magnitude.

S-X bond length: $SF_6 < SF_5Cl < SF_5Br$	(2 marks)
X-S-F bond angle: $SF_6 < SF_5Cl < SF_5Br$	(2 marks)
Total	(4 marks)

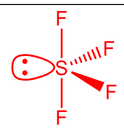
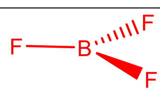
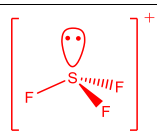
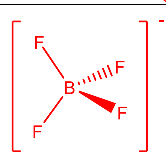
5.5 Sulfur tetrafluoride forms adducts with boron trifluoride which can be represented in ionic form as $[SF_3][BF_4]$. Write the corresponding chemical equation. In this reaction, indicate the compounds that exhibit Lewis acid-base properties, respectively. For the reactants and the ionic species of the adduct formed, write the chemical structure and electronic distribution of the central atom according to the hybridization theory developed by Pauling.

Chemical equation:

$SF_4(g) + BF_3(g) = [SF_3][BF_4](s)$

Lewis base   Lewis acid

(4 marks)

Chemical formula	Chemical structure	Electronic distribution
SF_4		S: [Ne] $3s^2\ 3p^2\ 3p^1\ 3p^1\ 3d^0\ 3d^0\ 3d^0\ 3d^0\ 3d^0$ S*: [Ne] $3s^2\ 3p^1\ 3p^1\ 3p^1\ 3d^1\ 3d^0\ 3d^0\ 3d^0\ 3d^0$ S: [Ne] $3(sp_3d)^2\ 3(sp_3d)^1\ 3(sp_3d)^1\ 3(sp_3d)^1\ 3(sp_3d)^1\ 3d^0\ 3d^0\ 3d^0\ 3d^0$ (2 marks)
BF_3	 (1 mark)	B: [He] $2s^2\ 2p^1\ 2p^0\ 2p^0$ B*: [He] $2s^1\ 2p^1\ 2p^1\ 2p^0$ B: [He] $2(sp_2)^1\ 2(sp_2)^1\ 2(sp_2)^1\ 2p^0$ (2 marks)
$[SF_3]^+$	 (1 mark)	S(1+): [Ne] $3s^2\ 3p^1\ 3p^1\ 3p^1\ 3d^0\ 3d^0\ 3d^0\ 3d^0\ 3d^0$ S(1+): [Ne] $3(sp_3)^2\ 3(sp_3)^1\ 3(sp_3)^1\ 3(sp_3)^1\ 3d^0\ 3d^0\ 3d^0\ 3d^0\ 3d^0$ (2 marks)
$[BF_4]^-$	 (1 mark)	B(1-): [He] $2s^2\ 2p^1\ 2p^1\ 2p^0$ B(1-)*: [He] $2s^1\ 2p^1\ 2p^1\ 2p^1$ B(1-)*: [He] $2(sp_3)^1\ 2(sp_3)^1\ 2(sp_3)^1\ 2(sp_3)^1$ (2 marks)
Total		(15 marks)

5.6 In addition to applications in the synthesis of organic compounds, thionyl chloride is used industrially to prepare anhydrous metal halides like $BiCl_3$. What is the simplified electronic distribution of bismuth? Explain why bismuth chlorides with higher oxidation states are not known.

Simplified electronic distribution

Bi: [Xe] $4f^{14}$ $5d^{10}$ $6s^2$ $6p^3$ (1 mark)

Of the five electrons that occupy the orbitals in the valence shell of the bismuth atom, two fill the *6s* orbital. According to the **inert pair effect**, the s orbital in the valence shell of the elements with the highest atomic number in groups 13, 14, and 15 of the periodic table, including bismuth (group 15, period 6), have extra stability, which discourages the participation of these electrons in chemical reactions. The existence of bismuth chlorides with oxidation states greater than three requires the participation of the electrons that occupy the *6s* orbital, which does not occur. (4 marks)

Total (5 marks)

Problem 6. Magnesium thermochemistry (20 points)

6.1	6.2	6.3	Total	Total
13	20	7	40	20

It is generally possible to consider that a given compound is thermodynamically unstable with respect to the decomposition into the constituent elements or to the disproportionation, if either of these chemical transformations involves a large release of energy.

6.1 Show by calculations that the unusual magnesium(I) fluoride is a thermodynamically unstable compound respect to its disproportionation at 298 K:

$2MgF(s) = Mg(s) + MgF_2(s)$

Note that the lattice energies of MgF_2 and MgF are 2952 and 930 kJ/mol, respectively. The sublimation heat of Mg(s) is 147.7 kJ/mol and the difference between the primary and secondary ionization energy of this metal is 713 kJ/mol. The above data were determined or estimated at STP.

$2\ MgF(s) = Mg(s) + MgF_2(s)$ $\Delta H_r^\emptyset$

$$\Delta H_r^\emptyset = \sum n(X)\Delta H_f^\emptyset(X)_{products} - \sum n(X)\Delta H_f^\emptyset(X)_{reactants}$$

$\Delta H_r^\emptyset = \Delta H_f^\emptyset(MgF_2) - 2\ \Delta H_f^\emptyset(MgF)$ (2 marks)

Applying the Born-Haber cycle to the formation of the MgF_2:

$\Delta H_f^\emptyset(MgF_2,\ s) = \Delta H_{subl}^\emptyset(Mg, s) + I_1(Mg, g) + I_2(Mg, g) + 2\ \Delta H_{dis}^\emptyset(F, g) + 2\ EA\ (F, g) - U(MgF_2, s)$

(3 marcas)

Applying the Born-Haber cycle to the formation of the MgF:

$\Delta H_f^\emptyset(MgF_2,\ s) = \Delta H_{subl}^\emptyset(Mg, s) + I_1(Mg, g) + \Delta H_{dis}^\emptyset(F, g) + EA(F, g) - U(MgF_2, s)$ (3 marks)

Substituting in the first equation:

$\Delta H_r^\emptyset = \Delta H_{subl}^\emptyset(Mg, s) + I_1(Mg, g) + I_2(Mg, g) + 2\ \Delta H_{dis}^\emptyset(F, g) + 2\ EA\ (F, g) - U\ (MgF_2, s) -$

$2\ [\Delta H_{subl}^\emptyset(Mg, s) + I_1(Mg, g) + \Delta H_{dis}^\emptyset(F, g) + EA\ (F, g)\ -\ U\ (MgF, s)]$

$\Delta H_r^\emptyset = -\ \Delta H_{subl}^\emptyset(Mg, s) - I_1(Mg, g) + I_2(Mg, g) -\ U\ (MgF_2, s) + 2\ U\ (MgF, s)$ (2 marks)

$\Delta H_r^\emptyset = -147.7\ kJ/mol + 713\ kJ/mol -\ 2952\ kJ/mol\ + 2\ * 930\ kJ/mol$

$\Delta H_r^\emptyset = -526.7\ kJ/mol$ (2 marks)

The disproportionation reaction of the unusual magnesium(I) fluoride involves the release of 526.7 kJ of energy, which is consistent with a thermodynamic instability of this compound respect to disproportionation. (1 mark)

Total (13 marks)

The formation energy of magnesium(II) fluoride was determined in 1964 by the direct reaction of the elements in a calorimeter. Combustion in fluorine atmosphere of a 0.50000 g sample of magnesium in the calorimeter causes the temperature to rise by 1.44093°C. The heat capacity of the calorimeter is 3433.46 cal/°C. However, as the reaction between both elements is very vigorous, a 195.8 g nickel container and 304.47 g of compacted magnesium(II) fluoride coating were also placed inside the calorimeter. After the calorimetric measurement, the sample under study was treated with a dilute solution of hydrochloric acid, producing 36.9 cm^3 of hydrogen under normal conditions.

6.2 What is the value of the formation energy of 1 mole of magnesium(II) fluoride?

$$Q_{sistema} = Q_{reaction} + Q_{calorim.} + Q_{Ni\ cruc.} + Q_{MgF_2\ coat.}$$
$$0 = Q_{reaction} + Q_{calorim.} + Q_{Ni\ cruc.} + Q_{MgF_2\ coat.}$$
$$Q_{reaction} = -Q_{calorim.} - Q_{Ni\ cruc.} - Q_{MgF_2\ coat.}$$ (2 marks)
$$Q_{reaction} = -\varepsilon_{calorim.}\ \Delta T - n(Ni)\ c_p(Ni)\Delta T - n(MgF_2)\ c_p(MgF_2)\ \Delta T$$
$$Q_{reaction} = -\varepsilon_{calorim.}\ \Delta T - \frac{m(Ni)}{M(Ni)}\ c_p(Ni)\Delta T - \frac{m(MgF_2)}{M(MgF_2)}\ c_p(MgF_2)\ \Delta T$$ (4 marks)
$$Q_{reaction} = -3433.46\ cal/°C * 1.44093°C - \frac{195.8\ g}{58.69\ g/mol} * 6.21\ cal/(mol \cdot °C) * 1.44093 - \frac{304.47\ g}{62.31 g/mol} * 14.72\ cal/(mol \cdot °C) * 1.44093°C$$ (2 marks)
$$Q_{reaction} = -5080.87\ cal * \frac{4{,}184\ J}{1\ cal}$$
$$Q_{reaction} = -21258.36\ J$$ (1 mark)

$Mg(s) + 2\ HCl(ac) = MgCl_2(ac) + H_2(g)$ (1 mark)
$$n(Mg)_{comb.} = n(Mg)_{sample} - n(Mg)_{not\ comb.}$$ (1 mark)
$$n(Mg)_{comb.} = \frac{m(Mg)_{sample}}{M(Mg)} - n(H_2)$$
$$n(Mg)_{comb.} = \frac{m(Mg)_{sample}}{M(Mg)} - \frac{V(H_2)}{V_m}$$ (3 marks)
$$n(Mg)_{comb.} = \frac{0.50000\ g}{24.31\ g/mol} - \frac{0.0369\ L}{22.4\ L/mol}$$
$$n(Mg)_{comb.} = 0.018920\ mol$$ (1 mark)

$Mg\ (s) + F_2\ (g) = MgF_2\ (s)$ (1 mark)
$$n(MgF_2) = n(Mg)_{comb.} = 0.018920\ mol$$ (1 mark)
$$\Delta H_f^{\emptyset}(MgF_2) = \frac{Q_{reaction}}{n(MgF_2)}$$ (1 mark)
$$\Delta H_f^{\emptyset}(MgF_2) = \frac{-21258.36\ J}{0.018920\ mol}$$
$$\Delta H_f^{\emptyset}(MgF_2) = -1123.59\ kJ/mol$$ (2 marks)

Total (20 marks)

6.3 Is magnesium(I) fluoride a thermodynamically stable salt respect to decomposition into its constituent elements? Use the information you obtained in the previous tasks.
If you could not solve parts 6.1 and 6.2, consider $\Delta H^o{}_f(MgF_2)$ = -1000 kJ/mol and ΔH^o(Reaction 1) = -500 kJ/mol

$MgF(s) = Mg(s) + 1/2\ F_2(g)$ $\Delta H^{\emptyset}_{r2} = -\Delta H^{\emptyset}_{f}(MgF)$ (2 marks)

$\Delta H^{\emptyset}_{r} = \Delta H^{\emptyset}_{f}(MgF_2) - 2\ \Delta H^{\emptyset}_{f}(MgF)$

$\Delta H^{\emptyset}_{f}(MgF) = \frac{\Delta H^{\emptyset}_{r} - \Delta H^{\emptyset}_{f}(MgF_2)}{2}$ (2 marks)

$\Delta H^{\emptyset}_{f}(MgF) = \frac{-526.7\ kJ/mol - (-1123.59\ kJ/mol)}{2}$

$\Delta H^{\emptyset}_{f}(MgF) = +\ 298.4\ kJ/mol$ (1 mark)

$\Delta H^{\emptyset}_{r2} = -\Delta H^{\emptyset}_{f}(MgF)$

$\Delta H^{\emptyset}_{r2} = -298.4\ kJ/mol$ (1 mark)

The decomposition reaction of the unusual magnesium(I) fluoride into its constituent elements involves the release of 298.4 kJ of energy, which predicts the thermodynamic instability of this compound respect to decomposition. (1 mark)

Total (7 marks)

Using the values of $\Delta H°_f(MgF_2)$ = -1000 kJ/mol and $\Delta H°$(Reaction 1) = -500 kJ/mol, a decomposition $\Delta H°$ value equal to -250 kJ/mol is obtained.

Data:

V_m = 22.4 L/mol in normal conditions (STP)

$c_p(Ni)$ = 6.21 cal/(mol·°C) and $c_p(MgF_2)$ = 14.72 cal/(mol·°C)

1 cal = 4.184 J

Problem 7. Hydrogen sulfide (21 points)

7.1	7.2	7.3	7.4	Total	Total
6	9	11	16	42	21

Hydrogen sulfide, H_2S, is a weak acid which can be used to precipitate metal ions from aqueous solutions by tight pH control.

7.1 Calculate the concentration of the sulfide anion in a saturated solution of hydrogen sulfide. Consider 0.1 mol of hydrogen sulfide dissolves in 1 L of water at STP. The primary and secondary dissociation constants of hydrogen sulfide are $1 \cdot 10^{-7}$ and $1 \cdot 10^{-14}$ respectively.

H_2S (ac) ⇌ HS^- (ac) + H^+ (ac) (1 mark)

HS^- (ac) ⇌ S^{2-} (ac) + H^+ (ac) (1 mark)

Since K_{a1} is greater than $1 \cdot 10^{-5}$ and the molar concentration of hydrogen sulfide greater than 0.001 mol/L, some approximations can be taken for the calculation:

$[H^+] \approx [HS^-]$ $\quad$ $[H_2S] \approx c(H_2S)$ (2 marks)

Considering this:

$K_{a2} = \frac{[H^+][S^{2-}]}{[HS^-]} \approx [S^{2-}] = 1 \cdot 10^{-14}$ (2 marks)

Total (6 marks)

7.2 Calculate the concentration of sulfide anion in 0.1 mol/L hydrochloric acid solution saturated with hydrogen sulfide.

$c(S^{2-}) = [S^{2-}] + [HS^-] + [H_2S] = [S^{2-}] + \frac{[H^+][S^{2-}]}{K_{a2}} + \frac{[H^+][HS^-]}{K_{a1}}$ (2 marks)

$c(S^{2-}) = [S^{2-}] + \frac{[H^+][S^{2-}]}{K_{a2}} + \frac{[H^+]^2[S^{2-}]}{K_{a1}*K_{a2}} = [S^{2-}]\left(1 + \frac{[H^+]}{K_{a2}} + \frac{[H^+]^2}{K_{a1}*K_{a2}}\right)$ (2 marks)

$c(S^{2-}) = [S^{2-}]\left(\frac{K_{a1}*K_{a2}+K_{a1}*[H^+]+[H^+]^2}{K_{a1}*K_{a2}}\right)$

$[S^{2-}] = c(S^{2-})\left(\frac{K_{a1}*K_{a2}}{K_{a1}*K_{a2}+K_{a1}*[H^+]+[H^+]^2}\right)$ (3 marks)

$[S^{2-}] = 0,1\left(\frac{1\cdot10^{-7}*1\cdot10^{-14}}{1\cdot10^{-7}*1\cdot10^{-14}+1\cdot10^{-7}*0,1+(0,1)^2}\right) = 1 \cdot 10^{-20}$ (2 marks)

Total (9 marks)

7.3 If 0.1 mol/L hydrochloric acid solution containing Hg^{2+} and Bi^{3+} ions at 5 mmol/L concentration is saturated with hydrogen sulfide, which sulfide precipitates first? What is the concentration of the cation involved after precipitation?

$K_{ps}(Bi_2S_3) = [Bi^{3+}]^2[S^{2-}]^3$ $\quad$ $K_{ps}(HgS) = [Hg^{2+}][S^{2-}]$ (2 marks)

To precipitate bismuth ion is needed a sulfide anion concentration of:

$[S^{2-}] = \sqrt[3]{\frac{K_{ps}(Bi_2S_3)}{[Bi^{3+}]^2}} = \sqrt[3]{\frac{1,6\cdot10^{-72}}{(5\cdot10^{-3})^2}} = 4,0 \cdot 10^{-23}$ (3 marks)

To precipitate the bismuth ion the sulfide anion concentration needed is:

$[S^{2-}] = \frac{K_{ps}(HgS)}{[Hg^{2+}]} = \frac{2 \cdot 10^{-32}}{5 \cdot 10^{-3}} = 4{,}0 \cdot 10^{-30}$ (2 marks)

Therefore, the first precipitate is HgS (1 mark)

And after saturation with H_2S, the concentration of S^{2-} will be practically equal to $1 \cdot 10^{-20}$, so the mercury ion concentration will be:

$[Hg^{2+}] = \frac{K_{ps}(HgS)}{[S^{2-}]} = \frac{2 \cdot 10^{-32}}{1 \cdot 10^{-20}} = 2{,}0 \cdot 10^{-12}$ (3 marks)

Total (11 marks)

7.4 Hg_2^{2+} ions present in the aqueous solutions of mercury(I) salts disproportionate easily. Determine by calculations if when saturated with hydrogen sulfide a solution of 0.1 mol/L of strong and soluble salt $Hg_2(NO_3)_2$ at pH = 1 also occurs the disproportion of the mercury(I) ions. Consider that the equation of the possible disproportion is:

$$Hg_2^{2+}(aq) + H_2S\ (aq) = Hg\ (s) + HgS\ (s) + 2\ H^+(aq)$$

The half-equations for disproportion of mercury(I) ions are:

$½Hg_2^{2+}(aq) + e^- \rightleftharpoons Hg\ (s)$ (1 mark)

$½Hg_2^{2+}(aq) \rightleftharpoons Hg^{2+}(ac) + e^-$ (1 mark)

The equation for the disproportion is:

$Hg_2^{2+}(aq) \rightleftharpoons Hg\ (s) + Hg^{2+}(aq)$ (1 mark)

And the precipitation equilibrium of mercury(II) ions with hydrogen sulfide:

$Hg^{2+}(aq) + H_2S\ (aq) \rightleftharpoons HgS\ (s) + 2\ H^+(aq)$ (2 marks)

Adding both equations obtains the equation given in the statement:

$Hg_2^{2+}(aq) + H_2S\ (aq) \rightleftharpoons Hg\ (s) + HgS\ (s) + 2\ H^+(aq)$ (1 mark)

The equilibrium constant for the second equation is:

$K_e = \frac{[H^+]^2}{[Hg^{2+}][H_2S]} = \frac{[H^+]^2[S^{2-}]}{[Hg^{2+}][H_2S][S^{2-}]} = \frac{K_{a1} * K_{a2}}{K_{ps}(HgS)} = \frac{1 \cdot 10^{-7} * 1 \cdot 10^{-14}}{2 \cdot 10^{-32}} = 5{,}0 \cdot 10^{10}$ (3 marks)

Taking this into account, it can be calculated the concentration of mercury(II) ions according to:

$[Hg^{2+}] = \frac{[H^+]^2}{K_e[H_2S]} = \frac{(0{,}1)^2}{5{,}0 \cdot 10^{10} * 0{,}1} = 2{,}0 \cdot 10^{-12}$ (2 marks)

The potential difference for the disproportionation equilibrium of mercury ions is:

$\Delta E = \Delta E° - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{[Hg_2^{2+}]}{[Hg^{2+}]} = \left(E^o_{Hg_2^{2+}/Hg} - E^o_{Hg^{2+}/Hg_2^{2+}}\right) - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{[Hg_2^{2+}]}{[Hg^{2+}]}$

$\Delta E = (0.80 - 0.91) - \frac{8.314 * 298.15}{1 * 96485} \cdot \ln \frac{0{,}1}{2.0 \cdot 10^{-12}} = 0.44\ V$ (4 marks)

Therefore, disproportionation of the mercury(I) ion occurs more easily in the presence of hydrogen sulfide. (1 mark)

Total (16 marks)

Data:

$K_{sp}(Bi_2S_3) = 1.6 \cdot 10^{-72}$ $K_{sp}(HgS) = 2.0 \cdot 10^{-32}$

Hg^{2+} — 0.91 V — Hg_2^{2+} — 0.80 V — Hg

| — 0.85 V — |

Problem 8. Electrolytic solutions (29 points)

8.1	8.2	8.3	8.4	8.5	8.6	8.7	8.8	Total	Total
4	3	7	5	2	4	2	2	29	29

Sodium hypochlorite is one of the most widely used substances as a water disinfectant and bleach for several centuries. In the process of hypochlorite production, first, by means of electrolysis of a brine solution (NaCl), gaseous chlorine was produced. For this, a current of 11.5 A was passed during 2 h and 50 min to a volume of 1 L.

8.1 State the possible cathodic and anodic half-reactions, as well as the global equation of the process that occurs.

Possible cathodic half-reaction:

$2\ H_2O + 2\ e^- = H_2\ (g) + 2\ OH^-\ (aq)$ (1 mark)

Possible anodic half-reactions:

$2\ Cl^-\ (aq) = Cl_2\ (g) + 2\ e^-$ (1 mark)

$2\ H_2O = O_2\ (g) + 4\ H^+\ (aq) + 4\ e^-$ (1 mark)

Global equations:

$2\ H_2O + 2\ Cl^-\ (aq) = H_2\ (g) + Cl_2\ (g) + 2\ OH^-\ (aq)$ (1 mark)

Total (4 marks)

8.2 How much amount of substance of the chlorine substance is produced?

$$n(Cl_2) = \frac{1}{2}n\left(\frac{Cl_2}{2}\right) = \frac{1}{2} * \frac{q}{F} = \frac{1}{2} * \frac{I*t}{F}$$ (1 mark)

$$n(Cl_2) = \frac{1}{2} * \frac{11.5\ A*10200\ s}{96485\ C} = 0.608\ mol$$ (2 marks)

Total (3 marks)

Chlorine that was obtained after the electrolysis was thoroughly bubbled into a 1 mol/L sodium hydroxide solution. The container where the solution was stored was hermetically closed after adding all the chlorine.

8.3 Show by calculations that the disproportionation of chlorine in a basic medium is a spontaneous process and state the global equation of this reaction.

The cathodic and anodic half-equations of this process are, respectively:		
½ Cl_2 (g) + e^- = Cl^- (aq)	$E°_{Cl_2/Cl^-}$ = 1.36 V	(1 mark)
½ Cl_2 (g) + 2 OH^- (aq) = ClO^- (aq) + H_2O + e^-	$E°_{ClO^-/Cl_2}$ = ?	(1 mark)
It is necessary to find the value of E_{ClO^-/Cl_2} to calculate the EMF of the cell		
$\Delta G°_{ClO^-/Cl_2} = \Delta G°_{ClO^-/Cl^-} - \Delta G°_{Cl_2/Cl^-}$		
$-FE°_{ClO^-/Cl_2} = -2FE°_{ClO^-/Cl^-} + FE°_{Cl_2/Cl^-}$		
$E°_{ClO^-/Cl_2} = 2E°_{ClO^-/Cl^-} - E°_{Cl_2/Cl^-} = 2 * 0{,}88 - 1{,}36 = 0{,}40\ \mathrm{V}$		(2 marks)
The global equation of the process is:		
Cl_2 (g) + 2 NaOH (aq) = NaCl (aq) + NaClO (aq) + H_2O		(1 mark)
$\Delta E° = E°_{Cl_2/Cl^-} - E°_{ClO^-/Cl_2}$ = 1.36 – 0.40 = 0.96 V		(1 mark)
$\Delta G° = -F\Delta E°$ = –96485 C/mol ∗ 0.96 V = –92.2 kJ/mol < 0		(1 mark)
Total		(7 marks)

8.4 Calculate the pH of the final solution. Note that the limiting substance is NaOH.

Cl_2 (g) + 2 NaOH (aq) = NaCl (aq) + NaClO (aq) + H_2O	
Since NaOH is the limiting substance, then:	
$\mathrm{c(NaClO)} = \frac{1}{2}\mathrm{c(NaOH)} = \frac{1}{2} * 1\ \mathrm{mol/L} = 0{,}5\ \mathrm{mol/L}$	(1 mark)
The salt formed (NaClO) produces hydrolysis in an aqueous medium:	
ClO^- (ac) + H_2O = HClO (ac) + OH^-	(1 mark)
$K_h = \frac{[OH^-][HClO]}{[ClO^-]} = \frac{[OH^-]^2}{[ClO^-]} = \frac{K_{H_2O}}{K_a(HClO)}$	(1 mark)
$[OH^-]^2 = \sqrt{\frac{K_{H_2O}}{K_a(HClO)} * [ClO^-]} = \sqrt{\frac{1\cdot 10^{-14}}{4\cdot 10^{-8}} * 0.5} = 3.54 \cdot 10^{-4}$	(1 mark)
$pH = 14 - pOH = 14 + \log[OH^-] = 14 + \log(3.54 \cdot 10^{-4}) = 10.55$	(1 mark)
Total	(5 marks)

From the brine solution (NaCl, 12.0% by mass) used as starting reagent, it is known that it has a density of 1.085 g/mL (at 25°C) and that when adding the salt to the water the volume of the latter increased 1.046 times.

8.5 Calculate the molar concentration of this solution.

$\mathrm{c(NaCl)} = \frac{\mathrm{n(NaCl)}}{\mathrm{V(D)}} = \frac{\mathrm{m(NaCl)}}{\mathrm{M(NaCl)*V(D)}} = \frac{\omega\mathrm{(NaCl)*m(D)}}{\mathrm{M(NaCl)*}\frac{\mathrm{m(D)}}{\rho\mathrm{(D)}}} = \frac{\omega\mathrm{(NaCl)}*\rho\mathrm{(D)}}{\mathrm{M(NaCl)}}$	(1 mark)
$\mathrm{c(NaCl)} = \frac{0.12*1086\ \mathrm{g/L}}{58.44\ \mathrm{g/mol}} = 2.23\ \mathrm{mol/L}$	(1 mark)
Total	(2 marks)

8.6 Calculate its molality. Consider the density of water at this temperature equal to 1 g/mL.

$b\left(\frac{NaCl}{H_2O}\right) = \frac{n(NaCl)}{m(H_2O)} = \frac{c(NaCl)*V(D)}{\rho(H_2O)*V(H_2O)}$	(2 marks)
Since adding salt to water increases its volume by 1.046 times, this translates into:	
$b\left(\frac{NaCl}{H_2O}\right) = \frac{c(NaCl)*1.046*V(H_2O)}{\rho(H_2O)*V(H_2O)} = \frac{1.046*c(NaCl)}{\rho(H_2O)}$	(1 mark)
$b\left(\frac{NaCl}{H_2O}\right) = \frac{1.046*2.23\ mol/L}{1\ kg/L} = 2.33\ mol/kg$	(1 mark)
Total	(4 marks)

8.7 What will be the melting point of this solution?

$\Delta T = K_c(H_2O) * b\left(\frac{NaCl}{H_2O}\right) = 1.86\ K \cdot kg \cdot mol^{-1} * 2.33\ mol/kg = 4.34\ K$	(1 mark)
$T\ =\ -4{,}34^oC$	(1 mark)
Total	(2 marks)

8.8 It was found that the solution begins to cool at -8.18°C. What is the reason for this difference between the calculated value in the previous section and the experimental one?

The difference is due to the dissociation of the salt (NaCl) into two ions (i ≈ 2)	(2 marks)
Total	(2 marks)

Data:

$K_c(H_2O) = 1.86\ K{\cdot}kg{\cdot}mol^{-1}$ $K_a(HClO) = 4.0{\cdot}10^{-8}$

$ClO_3^- \xrightarrow{0.33\ V} ClO_2^- \xrightarrow{0.67\ V} ClO^- \xrightarrow{?} Cl_2 \xrightarrow{1.36\ V} Cl^-$

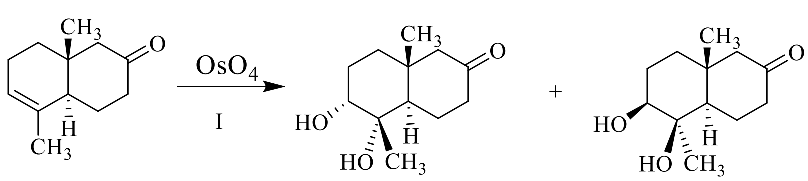
$ClO^- \xrightarrow{0.88\ V} Cl^-$

Problem 9. Cauthemone (19 points)

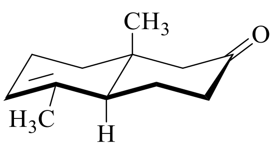
9.1	9.2	9.3	9.4	9.5	9.6	9.7	Total	Total
2	5	16	9	2	2	2	38	19

Cauthemone is a plant growth inhibitor isolated from the Mexican medicinal plant Cauthematl. It can be obtained through a sequence of non-enantioselective reactions, whereby in each synthesis step the optically active compounds are obtained as mixtures of enantiomers. To simplify the scheme only one of the enantiomers per compound is shown. Starting from compound **A**, the first step of the synthesis is shown below:

A **B** **C**

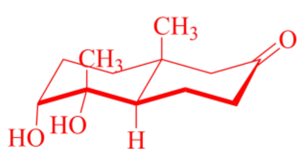


The three-dimensional structure of previous **A** stereoisomer can be represented in the chair form as follows:



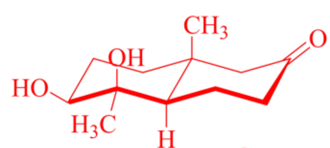
A

9.1 Illustrate the chair-shaped structure of **B** and **C**.



B

(1 mark)

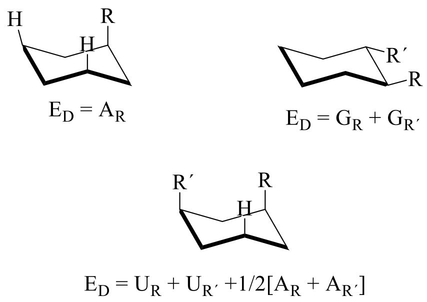


C

(1 mark)

Total (2 marks)

9.2 Justify by calculations which of the two diastereomers (**B** or **C**) is the most stable. Calculate the difference between the destabilization energies of diastereomers **B** and **C** using the data given in the following table. The destabilization energy (E_D) is the sum of the 1,3-diaxial interactions between a hydrogen atom and the R substituent (A_R), gauche interactions between the R and R´ substituents (G_R), and 1,3- diaxial between the R and R´ substituents (U_R).



A_R, G_R and U_R values of cyclohexane derivatives (kcal/mol)

R o R´	A_R	G_R	U_R
CH_3	1.8	0.9	1.8
CH_2R	1.8	0.4	1.8
OH	0.9	0.2	0.9

B: $E_D = 2\,U_{CH3} + A_{CH3} + A_{OH} + 2\,G_{OH} = 2 \times 1.8 + 1.8 + 0.9 + 2 \times 0.2 = 6.7$ kcal/mol (2 marks)

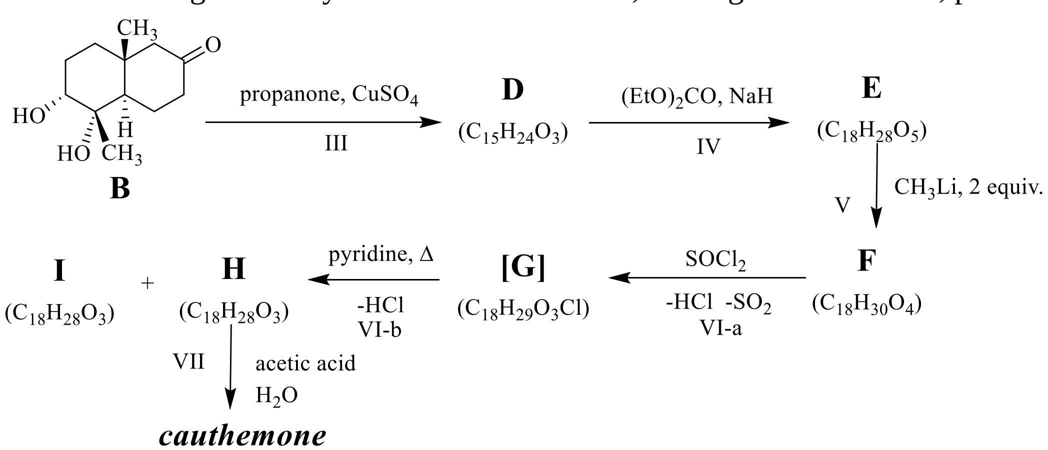
C: $E_D = U_{CH3} + U_{OH} + 1/2[A_{CH3} + A_{OH}] + 2\,G_{OH} + G_{CH3} + G_{OH} = 1.8 + 0.9 + 1/2[1.8 + 0.9] + 2 \times 0.2 + 0.9 + 0.2 = 5.55$ kcal/mol (2 marks)

$\Delta E_D = 6.7 – 5.55 = 1.15$ kcal/mol

It is verified by calculations that **C** is the most stable (lower energy) isomer. The destabilization energy of the isomer **B** is 1.15 kcal/mol higher than that of **C**. (1 mark)

Total (5 marks)

The second stage of the synthesis of ***cauthemone***, starting from isomer **B**, proceeds as follows:

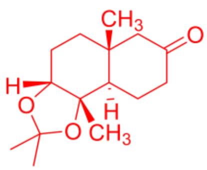


Steric effects determine that during the condensation of compound **D** with ethyl carbonate in the presence of sodium hydride, only the least hindered isomer is formed. The compound formed in the above reaction exists in the form of two tautomers, **T** and **E**, which are in equilibrium. Tautomer **E** possesses intramolecular hydrogen bonding and is more stable than its tautomer **T**. In fact, the reaction between **D** and ethyl carbonate first forms the **T** tautomer (not shown in the diagram) which rapidly transforms into **E**. Treatment of compound **F** with thionyl chloride produces an intermediate compound **G**, which in the presence of base rapidly transforms into compounds **H** and **I**, where **H** is the major product.

Catalytic hydrogenation of ***cauthemone*** produces a dehydro derivative, whereas only a compound **Y** ($C_{12}H_{18}O_4$) and propanone are obtained by reductive ozonolysis of this natural product. Treating ***cauthemone*** with 2,4-dinitrophenylhydrazine in acidic solution yields a product with a single hydrazone unit.

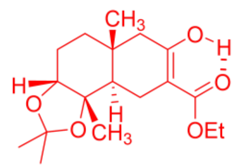
9.3 Write the three-dimensional structure of compounds **E** through **I**, **Y**, and ***cauhtemone***. Mark the stereogenic centers in the ***cauthemone*** structure and classify them according to the Cahn-Ingold-Prelog rules.

D



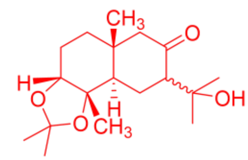
(1.5 marks)

E



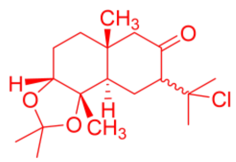
(1.5 marks)

F



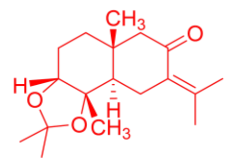
(1.5 marks)

G



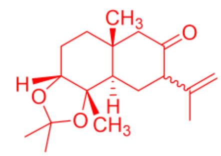
(1.5 marks)

H



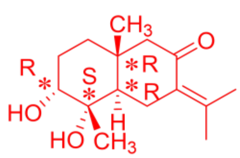
(1.5 marks)

I



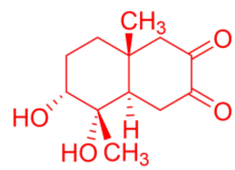
(1.5 marks)

Cauthemone



(1.5 marks) (+1 mark for each stereogenic center)

Y

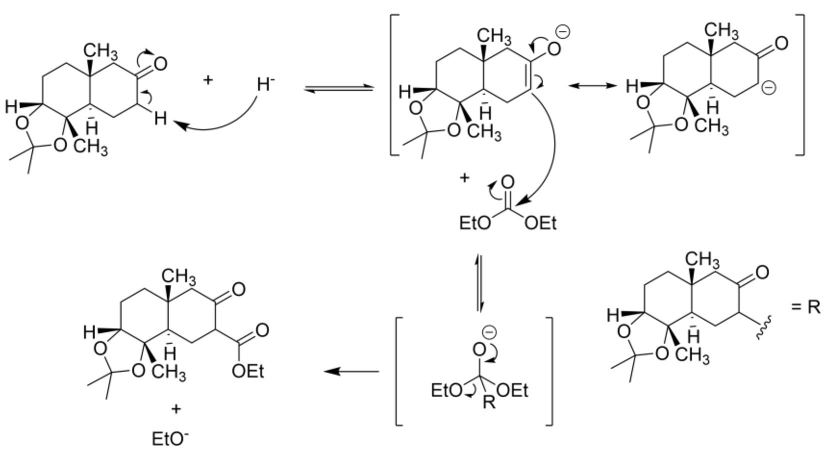


(1.5 marks)

Total (16 marks)

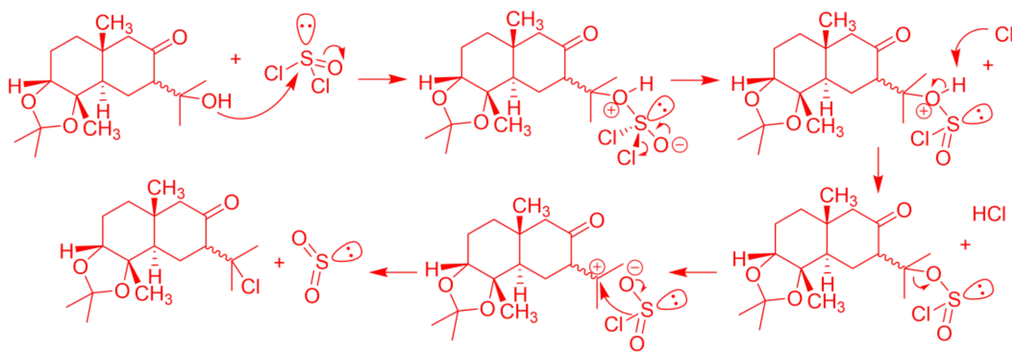
9.4 Write the mechanism of reactions IV, VI-a and VI-b.

Mechanism of reaction IV



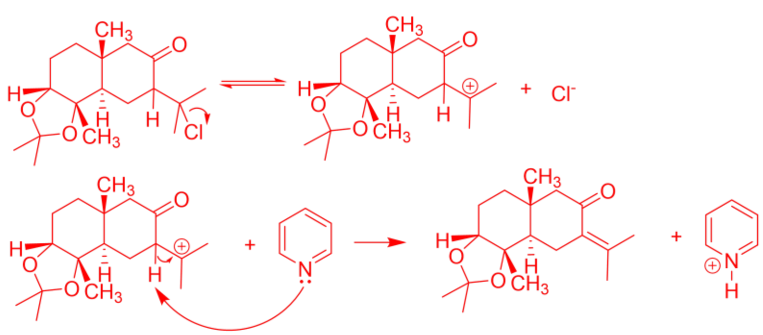
(3 marks)

Mechanism of reaction VI-a



(3 marks)

Mechanism of reaction VI-b



(3 marks)

9.5 Represent the tautomeric equilibrium between **E** and **T**.

Tautomeric equilibrium between E y T

(2 marks)

Total (2 marks)

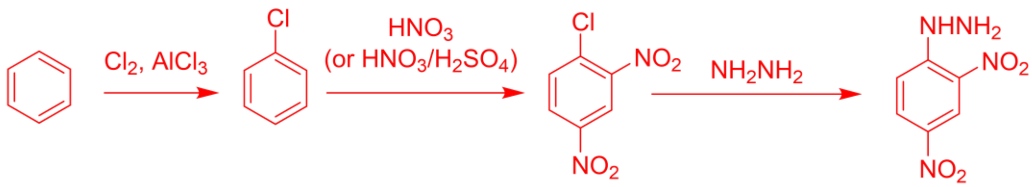
9.6 Why is **H** the major product in reaction VI-b?

According to Zaytzeff's rule, the major product of E1 elimination reaction is the one with the most substituted double bond, which is related to the stability of unsaturated compounds. In addition, the removal of the alpha proton from the ketone in the second step of the elimination reaction is more favored because it has a higher acidic character and the product is a more stable conjugated ketone.

(2 marks)

Total (2 marks)

9.7 State a reasonable sequence of steps for the synthesis of the reagent 2,4-dinitrophenylhydrazine from benzene.



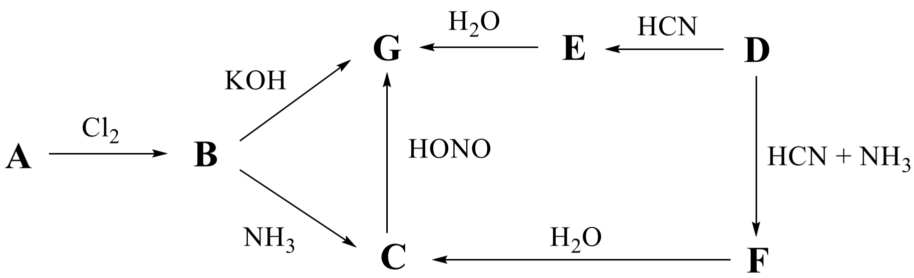
(2 marks)

Total (2 marks)

Problem 10. Organic compounds (12 points)

10.1	10.2	Total	Total
8	16	24	12

Substance **G** can be prepared by various methods, according to the following scheme:



Compound **A** has 48.60% of carbon, 8.10% of hydrogen, and 43.30% of oxygen, in mass percentage. When performing the volumetric titration of 0.222 g of **A** dissolved in water, 15.0 mL of 0.2 mol/L NaOH were needed for its total neutralization.

Compound **D** contains 54.54% of carbon, 9.09% of hydrogen and 36.37% of oxygen. It combines equimolarly with $NaHSO_3$ to give a compound with 21.6% of sulfur, in mass percentage.

10.1 Write the empirical formula of **A** and **D**.

In 100 g of **A**, 48.6 g of carbon, 8.1 g of hydrogen, and 43.3 g of oxygen are present. This corresponds to the amounts of substance:

$n(C) = \frac{48.6\ g}{12.01\ g/mol} = 4.05\ mol$ $n(H) = \frac{8.1\ g}{1.008\ g/mol} = 8.04\ mol$

$n(O) = \frac{43.3\ g}{16.00\ g/mol} = 2.71\ mol$ (2 marks)

Dividing all the quantities by the smallest value (2.71), we have:

C: $\frac{4.05}{2.71} = 1.5$ H: $\frac{8.04}{2.71} = 3$ O: $\frac{2.71}{2.71} = 1$ (1 mark)

Multiplying by 2 to have integer values gives the empirical formula of **A**: $C_3H_6O_2$.

(1 mark)

In 100 g of **D**, 54.54 g of carbon, 9.09 g of hydrogen, and 36.37 g of oxygen are present. This corresponds to the amounts of substance:

$n(C) = \frac{54.54\ g}{12.01\ g/mol} = 4.54\ mol$ $n(H) = \frac{9.09\ g}{1.008\ g/mol} = 9.02\ mol$

$n(O) = \frac{36.37\ g}{16.00\ g/mol} = 2.27\ mol$ (2 marks)

Dividing all the quantities by the smallest value (2.27), we have:

C: $\frac{4.54}{2.27} = 2$ H: $\frac{9.02}{2.27} = 4$ O: $\frac{2.27}{2.27} = 1$ (1 mark)

Therefore, the empirical formula of **D** is: C_2H_4O. (1 mark)

Total (8 marks)

10.2 Deduce the structures of the compounds from **A** to **G**.

From the qualitative test with $NaHSO_3$, the presence of a carbonyl group in compound **D** can be deduced. As the reaction with $NaHSO_3$ is addition, the molar mass of the compound obtained at the end of the reaction is the sum of both molar masses. From the percent by mass of sulfur it follows that:

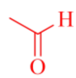
$$\omega(\mathrm{S}) = \frac{\mathrm{M(S)}}{\mathrm{M}(\mathbf{D}+\mathrm{NaHSO_3})}$$

$$\mathrm{M}(\mathbf{D} + \mathrm{NaHSO_3}) = \frac{32{,}06\ \mathrm{g/mol}}{0{,}216} = 148{,}4\ \mathrm{g/mol}$$

Subtracting the molar mass of $NaHSO_3$ gives the molar mass of **D**:

$$M(D) = 148.4\ \mathrm{g/mol} - (32.06 + 22.99 + 1.008 + 16 * 3)\mathrm{g/mol} = 44.3\ \mathrm{g/mol}$$

Based on the empirical formula C_2H_4O, which has a molar mass of 44.1 g/mol, it follows that the molecular formula matches the empirical one. According to this, the only possible structure for D is:

D 

(3 marks)

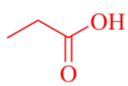
From the quantitative test with NaOH it can be deduced that compound **A** behaves like an acid in solution, and that:

$$\mathrm{n(H^+)}_{\mathbf{A}} = \mathrm{n(NaOH)} = \mathrm{c(NaOH)} * \mathrm{V_{bur}} = 0.2\ \mathrm{mol/L} * 0.015\ \mathrm{L} = 0.003\ \mathrm{mol}$$

Assuming one acid group for each molecule of **A**, then the molar mass of **A** is:

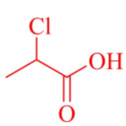
$$\mathrm{M}(\mathbf{A}) = \frac{0.222\ \mathrm{g}}{0.003\ \mathrm{mol}} = 74\ \mathrm{g/mol}$$

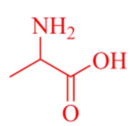
Based on the empirical formula $C_3H_6O_2$, which has a molar mass of 74.1 g/mol, it follows that the molecular formula matches the empirical one. Therefore, the structure of this monocarboxylic acid is:

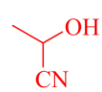
A 

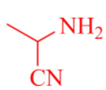
(3 marks)

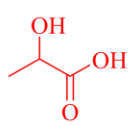
The structures of the other compounds are:

B 

C 

E 

F 

G 

(2 marks each one)

Total (16 marks)